

Project Initialization and Planning Phase

Date	1 July 2026
Team ID	
Project Title	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed project aims to improve flood prediction accuracy using machine learning techniques by analyzing historical weather patterns, rainfall intensity, water level variations, and environmental conditions. It helps authorities provide early warnings and reduce damage caused by floods. The system focuses on predictive analytics for disaster management and public safety.

Project Overview	
Objective	The primary objective is to develop a machine learning-based flood prediction system that can forecast flood occurrences early and accurately, enabling better disaster preparedness and reducing risks to life and property.
Scope	The project analyzes environmental and weather-related data such as rainfall, river water levels, temperature, and humidity to predict possible flood situations using intelligent prediction models.
Problem Statement	
Description	Traditional flood warning systems often depend on manual monitoring and limited forecasting methods, resulting in delayed warnings and inaccurate predictions.
Impact	Poor flood prediction can lead to severe property damage, economic loss, infrastructure destruction, and threats to human life in flood-prone areas.
Proposed Solution	
Approach	Using machine learning algorithms to process historical flood and weather datasets, identify patterns, and predict future flood risks in real time.
Key Features	• Early flood prediction using machine learning models. • Real-time environmental data analysis. • Better disaster management planning. • Improved prediction accuracy compared to traditional systems.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU processing requirements	Intel i5 / GPU support
Memory	RAM specifications	8 GB RAM
Storage	Space for datasets and models	500 GB SSD
Software		
Frameworks	Machine Learning framework	Python, Flask
Libraries	Data science libraries	Scikit-learn, Pandas, NumPy, Matplotlib
Development Environment	IDE	Jupyter Notebook, PyCharm
Data		
Data	Historical weather and flood data	Kaggle Dataset, CSV files